

1) Dom. Rec. Asinto, Interc. GP

$$f(x) = \log_3(5-x) + 4$$

$$\text{Dom } f(x): \{x \in \mathbb{R} : \log_3(5-x) \in \mathbb{R}\} : 5-x > 0 ; x < 5 \Rightarrow \text{Dom } f(x) =]-\infty, 5[$$

$$\text{Rec: } \{y \in \mathbb{R}, x \in]-\infty, 5[: f(x) = \log_3(5-x) + 4\}$$

$$y = \log_3(5-x) + 4$$

$$y - 4 = \log_3(5-x) \quad (/ 0^{'})$$

$$3^{y-4} = 5-x$$

$$x = 5 - 3^{y-4}$$

$$\text{Rec: } \mathbb{R}$$

Asintota: Cuando $x=5$, se indetermina

$$y=0 \Rightarrow \log_3(5-x) + 4 = 0$$

$$\log_3(5-x) = -4 \quad (/ 3^{'})$$

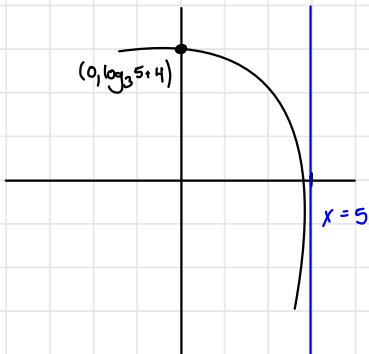
$$5-x = 3^{-4}$$

$$5 - 3^{-4} = x \quad (5 - 3^{-4}, 0)$$

$$5 - \frac{1}{81} \quad (+)$$

$$x=0 \Rightarrow y = \log_3(5) + 4 \quad (0, \log_3(5) + 4)$$

GP:



$$2) \frac{z^x + 4x - 2}{x+1} > 4$$

$$\frac{z^x + 4x - 2 - 4x - 4}{x+1} > 0$$

$$\frac{z^x - 6}{x+1} > 0$$

$$z^x - 6 = 0$$

$$z^x = 6 \quad / \log_z$$

$$x = \log_z(6)$$

$$\begin{array}{c|ccc} z^x - 6 & -\infty & -1 & \log_z 6 & +\infty \\ \hline & - & - & 0 & + \\ x+1 & + & 0 & + & + \\ & + & - & + & \end{array} \quad S:]-\infty, -1[\cup]\log_z 6, +\infty[$$